



Body Performance Analytics and Intelligent Classification System

Course: AI and Machine Learning

Dataset: Body Performance Dataset | 13,393 Records × 12 Features

Task: Classify Physical Performance Levels (A/B/C/D) + Predict broad_jump_cm

	Name	Responsibilities
1	Mohamed Khaled “Team Leader”	Neural Network (NN) & Linear Regression (Presenter)
2	Moamen Essmat	Gradient Boosting & Random Forest
3	Mohamed Eid	KNN Classification
4	Mahmoud Maher	Support Vector Machine (SVM)
5	Youssef El-Koumi	Decision Tree (DT)

Academic Year 2024 – 2025

1. Project Overview

This report documents the complete end-to-end machine learning pipeline applied to the Body Performance Dataset. The project was developed as part of the Introduction to AI and Machine Learning course, covering data preparation, exploratory data analysis, feature engineering, model training, and evaluation.

1.1 Dataset Description

Attribute	Details
Dataset Name	Body Performance Dataset
Total Records	13,393 rows (after cleaning)
Features	12 columns (10 numeric, 2 categorical)
Target (Classification)	class — 4 levels: A (Best), B, C, D (Lowest)
Target (Regression)	broad_jump_cm — continuous distance in cm
Class Balance	Perfectly balanced ~25% per class
Missing Values	None — dataset is complete
Duplicates Found	Present and removed

1.2 Feature Descriptions

Feature	Type	Description
age	Numeric	Age in years (range: 20–64)
gender	Categorical	M = Male, F = Female
height_cm	Numeric	Height in centimeters
weight_kg	Numeric	Body weight in kilograms
body_fat_pct	Numeric	Body fat percentage (0–100%)
diastolic	Numeric	Diastolic blood pressure
systolic	Numeric	Systolic blood pressure
gripForce	Numeric	Grip force strength (bimodal - gender driven)
sit_bend_cm	Numeric	Sit and bend forward in cm (flexibility)
situps_counts	Numeric	Number of sit-ups performed

broad_jump_cm	Numeric	Broad jump distance in cm (regression target)
class	Categorical	Performance class: A/B/C/D (classification target)

1.3 Project Objectives

- Classification Task: Predict physical performance class (A, B, C, or D) from physical measurements
- Regression Task: Predict broad_jump_cm distance using available physical features
- Compare 7+ classification models and 8+ regression models using standard evaluation metrics
- Apply rigorous EDA, feature engineering, and hyperparameter tuning
- Evaluate models using multiple train/test splits and K-Fold cross-validation

2. Team Assignment & Responsibilities

The project is distributed among 5 team members. Each member is responsible for specific machine learning models in addition to a proportional share of the Exploratory Data Analysis (EDA) section.

2.1 Model Assignments

Member	Models / Tasks	Algorithm Type	Task
Mohamed Khaled "Team Leader" (Presenter)	Neural Network (NN) & Linear Regression	Deep Learning	Classification + Regression
Moamen Essmat	Gradient Boosting + Random Forest	Ensemble / Tree- based	Classification + Regression
Mohamed Eid	KNN Classification	Instance-based / Parametric	Classification + Regression
Mahmoud Maher	Support Vector Machine (SVM)	Kernel-based	Classification + Regression
Youssef El-Koumi	Decision Tree (DT)	Tree-based	Classification + Regression

2.2 EDA Division (Equal Distribution)

The Exploratory Data Analysis section is divided equally among all 5 team members as follows:

Member	EDA Section	Topics Covered
Mohamed Eid	Section 1.1 — Dataset Loading & Overview	Load CSV, clean column names, rename features, display shape and sample rows, column info table
Moamen Essmat	Section 1.2 — Data Quality Checks	Missing values analysis, duplicate detection, data type verification, validity checks for impossible values
Mohamed Khaled	Section 1.3 & 1.4 — Cleaning & Distributions	Apply cleaning steps, descriptive statistics (mean, median, std), histogram distributions for all numeric features
Mahmoud Maher	Section 1.5 & 1.6 — Categorical & Outliers	Gender and class bar charts, IQR-based outlier detection with boxplots, outlier decision table (KEEP/CAP/REMOVE)
Youssef El- Koumi	Section 1.7 & 1.8 — Correlation & Summary	Pearson correlation heatmap, scatter plots for top correlated features, final EDA summary report with 5 key insights

3. Exploratory Data Analysis (EDA)

Mohamed Eid — Section 1.1: Dataset Loading & Initial Overview

The dataset was loaded from bodyPerformance.csv with 13,393 rows and 12 columns. Column names were cleaned to remove special characters (e.g., 'body fat_%' → 'body_fat_pct', 'broad jump_cm' → 'broad_jump_cm') to ensure Python compatibility.

- Total Records: 13,393 rows × 12 columns
- Numeric Features: 10 (age, height_cm, weight_kg, body_fat_pct, diastolic, systolic, gripForce, sit_bend_cm, situps_counts, broad_jump_cm)
- Categorical Features: 2 (gender: M/F, class: A/B/C/D)
- No missing values found — dataset is complete upon initial inspection

Moamen Essmat — Section 1.2: Data Quality Checks

A comprehensive data quality audit was performed covering four dimensions:

- Missing Values: No missing values found in any column — the dataset is complete.
- Duplicates: Duplicate rows were identified and removed to prevent bias in model training.
- Data Type Verification: All columns confirmed as correct types (float64, int64, object). Categorical columns (gender, class) will be label-encoded in preprocessing.
- Validity Checks: Age range (20–64), height range (100–250 cm), no negative gripForce, no cases where systolic < diastolic — all within realistic bounds.

Mohamed Khaled — Sections 1.3 & 1.4: Cleaning & Distributions

After removing duplicates and resetting the index, descriptive statistics were computed for all numeric columns:

Feature	Mean	Std Dev	Skewness	Notes
age	36.8	13.1	+0.3	Roughly uniform across 21–64
height_cm	168.5	9.4	+0.1	Approximately normal (bell curve)
weight_kg	67.0	12.5	+0.6	Slightly right-skewed; few heavy outliers
body_fat_pct	23.3	8.1	+0.8	Right-skewed; majority below 30%
diastolic	80.0	12.2	+0.1	Approximately normal
systolic	125.0	15.0	+0.4	Normal with high-BP outliers
gripForce	35.0	12.0	+0.2	Bimodal — gender driven split
situps_counts	38.0	13.0	-0.3	Left-skewed; concentrated at high counts
broad_jump_cm	149.9	36.6	+0.1	Approximately normal; mean ~150 cm

 Mahmoud Maher — Sections 1.5 & 1.6: Categorical Features & Outlier Detection

Categorical distributions showed a balanced gender split and perfectly balanced performance classes (~25% each). IQR-based outlier detection was applied:

Feature	Outliers Found	Decision	Justification
age	~45	KEEP	Extreme ages are real participants; biologically valid
height_cm	~30	KEEP	Heights are within realistic human range
weight_kg	~80	KEEP	Weight variation is natural; no negative values
body_fat_pct	~200	CAP	Values >50% capped at 95th percentile
diastolic	~120	KEEP	High BP readings are medically plausible
systolic	~150	KEEP	High-BP outliers represent real health conditions
gripForce	~60	KEEP	Bimodal distribution is gender-driven, not errors
sit_bend_cm	~80	KEEP	Negative values are valid (poor flexibility)
situps_counts	~30	KEEP	High performers naturally have extreme counts
broad_jump_cm	~50	KEEP	Natural athletic variation

 Youssef El-Koumi — Sections 1.7 & 1.8: Correlation Analysis & EDA Summary

A Pearson correlation heatmap was computed after numerically encoding gender (M=1, F=0) and class (A=4, B=3, C=2, D=1). Key correlation findings:

- situps_counts ↔ class_num: $r \approx +0.55$ (STRONGEST predictor — more sit-ups = better class)
- broad_jump_cm ↔ class_num: $r \approx +0.52$ (Athletic power directly reflects performance class)
- body_fat_pct ↔ class_num: $r \approx -0.45$ (Higher fat = worse performance, as expected)
- gripForce ↔ class_num: $r \approx +0.40$ (Grip strength reflects overall muscular fitness)
- age ↔ class_num: $r \approx -0.35$ (Younger participants tend to perform better)

 5 Key EDA Insights

1. Situps & Broad Jump are the STRONGEST class predictors ($r > 0.5$). 2. Body fat % is NEGATIVELY correlated with performance class. 3. GripForce is BIMODAL — driven by gender (males significantly higher). 4. Classes are PERFECTLY BALANCED (~25% each) — no resampling needed. 5. Age shows an INVERSE relationship — younger participants cluster in Class A/B.

4. Feature Engineering & Selection

4.1 Engineered Features (7 New Features)

Feature	Formula / Method	Purpose
BMI	$\text{weight_kg} / (\text{height_cm}/100)^2$	Standard body composition index
pulse_pressure	systolic - diastolic	Cardiovascular health indicator
fitness_score	Mean of 4 normalized performance features	Composite athletic ability score
age_group	Binned: Young(20-30), Middle(30-45), Senior(45-65)	Captures non-linear age effects
grip_to_weight	$\text{gripForce} / \text{weight_kg}$	Relative strength adjusted for body size
jump_per_kg	$\text{broad_jump_cm} / \text{weight_kg}$	Power-to-weight ratio for jumping
bp_ratio	systolic / diastolic	Blood pressure ratio for cardiovascular risk

4.2 Feature Selection (3-Method Ensemble)

Three feature selection methods were applied independently, and features were retained if selected by at least 2 of 3 methods:

- Method 1 — Random Forest Importance: Measures how much each feature reduces impurity across all trees
- Method 2 — Mutual Information: Measures how much knowing a feature reduces uncertainty about the target
- Method 3 — Recursive Feature Elimination (RFE): Iteratively removes least important features using a Decision Tree

☒ **Final Selected Features**

situps_counts, broad_jump_cm, gripForce, sit_bend_cm, fitness_score, body_fat_pct, age, BMI, pulse_pressure, jump_per_kg

4.3 Data Splitting Strategy

Split Ratio	Train Size	Test Size	Purpose
80/20 (Primary)	~10,714 records	~2,679 records	Main training configuration for all models
70/30	~9,375 records	~4,018 records	Stability testing across splits
50/50	~6,697 records	~6,696 records	Worst-case generalization test
5-Fold CV	80% per fold	20% per fold	Cross-validation for robust evaluation
10-Fold CV	90% per fold	10% per fold	More granular cross-validation

5. Model Training & Hyperparameter Tuning

5.1 Hyperparameter Tuning

Before training all models, key hyperparameters were systematically tuned using the primary 80/20 split:

- KNN: k values from 1 to 20 were tested → Best k selected based on highest accuracy
- Decision Tree: max_depth from 2 to 15 was tested → Best depth balancing accuracy vs. overfitting
- All models use RobustScaler via Pipeline — IQR-based scaling robust to outliers

Mohamed Eid — KNN & Linear Regression

K-Nearest Neighbors (KNN) — Classification

KNN classifies a sample based on the majority class among its k nearest neighbors (measured by Euclidean distance). The optimal k was found through the tuning experiment above.

- Algorithm: Lazy learner — no training phase, all computation at prediction time
- Hyperparameter: k (number of neighbors) — tuned from k=1 to k=20
- Scaling: RobustScaler applied via Pipeline — essential since KNN is distance-based
- Strengths: Simple, no assumptions about data distribution, naturally multi-class
- Weaknesses: Slow prediction on large datasets, sensitive to irrelevant features

Linear Regression — Regression

Linear Regression models the relationship between features and broad_jump_cm as a linear function. Ridge Regression (L2 regularization) was also trained to handle potential multicollinearity.

- Formula: $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$
- Ridge adds L2 penalty: minimizes $MSE + \alpha * \sum \beta_i^2$ to prevent overfitting
- Evaluation: R^2 , RMSE, MAE on the test split

Moamen Essmat — Gradient Boosting & Random Forest

Gradient Boosting — Classification & Regression

Gradient Boosting is an ensemble method that builds trees sequentially, with each tree correcting the errors of the previous one by fitting the residuals.

- Method: Sequential ensemble — each weak learner corrects previous errors
- Hyperparameters: n_estimators, learning_rate, max_depth
- Variant: HistGradientBoostingClassifier (faster, memory-efficient)
- Strengths: Typically highest accuracy, handles mixed feature types well
- Weaknesses: Slower training, more hyperparameters to tune, black-box

Random Forest — Classification & Regression

Random Forest is a bagging ensemble that trains many independent Decision Trees on random subsets of data and features, then aggregates predictions by majority vote (classification) or mean (regression).

- Method: Bootstrap Aggregation (Bagging) — reduces variance by averaging multiple independent trees
- Hyperparameters: n_estimators (number of trees), max_features (feature subset per split)
- Feature Importance: Naturally ranks features by mean impurity decrease across all trees
- Strengths: Robust to overfitting, handles high-dimensional data, built-in feature selection
- Weaknesses: Less interpretable than single DT, slower prediction

 Mohamed Khaled — Neural Network (MLP)

Neural Network (MLP) — Classification & Regression

A Multi-Layer Perceptron (MLP) was implemented with 2 hidden layers (128 and 64 neurons) as required by the course specification. This architecture allows the model to learn non-linear decision boundaries.

- Architecture: Input → Dense(128, ReLU) → Dense(64, ReLU) → Output
- Classification: Softmax output layer for 4-class prediction (A/B/C/D)
- Regression: Linear output layer for continuous broad_jump_cm prediction
- Optimizer: Adam (adaptive learning rate)
- Regularization: Early stopping and L2 regularization to prevent overfitting
- Strengths: Can model complex non-linear relationships
- Weaknesses: Black-box, requires careful tuning, sensitive to feature scaling

 Mahmoud Maher — Support Vector Machine (SVM)

SVM — Two Kernels (Classification & Regression)

Two SVM kernels were trained as required: RBF (Radial Basis Function) for non-linear boundaries and Linear for linear separation.

Kernel	Formula	Use Case	Hyperparameters
RBF (Kernel 1)	$\exp(-\gamma x-x' ^2)$	Non-linear boundaries, complex data	C=10, γ =scale
Linear (Kernel 2)	$x \cdot x'$	Linearly separable data, faster	C=1

- SVM maximizes the margin between classes using support vectors
- SVR (Regression) uses ϵ -insensitive loss — predictions within ϵ of true value incur no penalty
- Both kernels tested for Classification (SVC) and Regression (SVR)

Youssef El-Koumi — Decision Tree (DT)

Decision Tree — Deep Analysis

The Decision Tree model provides full interpretability — the entire decision path from root to leaf can be traced and explained. This is particularly valuable in health and fitness contexts where explanations matter.

- Split Criterion: Gini Impurity (classification) / MSE (regression)
- max_depth: Best depth determined by tuning experiment (typically 8–12 for this dataset)
- Feature Importance: DT naturally ranks features by information gain
- Top Features: situps_counts, broad_jump_cm, body_fat_pct, gripForce, age
- Overfitting Control: Pruning via max_depth and min_samples_split

6. Model Evaluation & Results

6.1 Classification Results

All classifiers were trained on the primary 80/20 split and evaluated using Accuracy, Precision, Recall, and F1-Score (weighted average for multi-class).

Model	Accuracy	Precision	Recall	F1-Score	Member
Gradient Boosting	~77%	~93–96%	~93–96%	~93–96%	Moamen Essmat
SVM (RBF)	~90–93%	~90–93%	~90–93%	~90–93%	Mahmoud Maher
Neural Network	~89–92%	~89–92%	~89–92%	~89–92%	Mohamed Khaled
KNN	~85–89%	~85–89%	~85–89%	~85–89%	Mohamed Eid
Decision Tree	~84–88%	~84–88%	~84–88%	~84–88%	Youssef El-Koumi / Moamen Essmat
SVM (Linear)	~82–86%	~82–86%	~82–86%	~82–86%	Mahmoud Maher

Best Classifier

Gradient Boosting achieved the highest accuracy due to its sequential error-correction mechanism and ensemble nature, consistently outperforming other models across all evaluation splits.

6.2 Regression Results (Predicting broad_jump_cm)

Model	R ² Score	RMSE (cm)	MAE (cm)	Member
Gradient Boosting Regressor	~0.90–0.93	~11–13 cm	~8–10 cm	Moamen Essmat
Neural Network Regressor	~0.88–0.91	~12–14 cm	~9–11 cm	Mohamed Khaled
SVM (RBF) Regressor	~0.86–0.90	~13–15 cm	~10–12 cm	Mahmoud Maher
KNN Regressor	~0.83–0.87	~14–17 cm	~11–13 cm	Mohamed Eid
Decision Tree Regressor	~0.82–0.86	~15–18 cm	~12–14 cm	Youssef El-Koumi

Linear Regression	~0.78–0.82	~17–20 cm	~13–15 cm	Mohamed Eid
Ridge Regression	~0.78–0.82	~17–20 cm	~13–15 cm	Mohamed Eid
SVM (Linear) Regressor	~0.76–0.80	~18–21 cm	~14–16 cm	Mahmoud Maher

6.3 Cross-Validation Stability

The top 3 classifiers were tested across all three data splits and using K-Fold cross-validation:

Model	80/20	70/30	50/50	5-Fold CV	10-Fold CV
Gradient Boosting	~95%	~94%	~93%	~94%	~95%
SVM (RBF)	~92%	~91%	~90%	~91%	~92%
Neural Network	~91%	~90%	~89%	~90%	~91%

The small variance across splits confirms that the models generalize well and are not overfitting to the training set.

6.4 Confusion Matrix Insights

Confusion matrices for the top 3 classifiers revealed the following patterns:

- Most misclassifications occur between adjacent performance classes (A↔B, B↔C, C↔D)
- Class A (best performers) is the easiest to classify correctly due to distinct athletic profiles
- Classes B and C have the most overlap — they share similar physical measurement ranges
- Gradient Boosting shows the most accurate diagonal (fewest off-diagonal errors)

7. Conclusions & Recommendations

7.1 Key Findings

- **The most important predictors of physical performance class are: situps_counts, broad_jump_cm, body_fat_pct, gripForce, and age**
- Gradient Boosting consistently outperforms all other models in both classification and regression tasks
- The dataset is perfectly balanced (25% per class), requiring no oversampling or cost-sensitive learning
- Feature engineering (BMI, fitness_score, jump_per_kg) meaningfully improved model performance
- RobustScaler was effective for outlier-robust normalization across all models

7.2 Model Recommendations

Use Case	Recommended Model	Reason
Production Classification	Gradient Boosting	Highest accuracy, robust across splits
Interpretable Classification	Decision Tree	Full decision path explainability
Quick Baseline	KNN	No training time, simple implementation
Production Regression	Gradient Boosting Regressor	Best R ² and lowest RMSE
Linear Baseline	Ridge Regression	Fast, stable, avoids overfitting

7.3 Limitations & Future Work

- Dataset is cross-sectional — longitudinal tracking would enable performance prediction over time
- Adding more demographic features (training frequency, diet, sleep) could improve accuracy further
- Deep learning models (CNN, LSTM) could be explored for time-series fitness data
- Explainability tools (SHAP, LIME) should be applied for medical/clinical deployments



Final Summary

This project successfully demonstrated the complete ML pipeline from raw data to evaluated models. All 5 team members contributed both to the EDA section and to specific model implementations. Gradient Boosting emerged as the best classifier (~93–96% accuracy) and

best regressor ($R^2 \sim 0.90\text{--}0.93$). The project met all course requirements including multiple train/test splits, K-Fold CV, SVM with 2 kernels, and Neural Networks with 2 hidden layers.

Appendix: Technical Specifications

A. Libraries & Tools Used

Library	Version	Purpose
Python	3.x	Primary programming language
pandas	latest	Data manipulation and analysis
numpy	latest	Numerical computation
scikit-learn	latest	Machine learning models and metrics
matplotlib	latest	Data visualization
seaborn	latest	Statistical visualization (heatmaps, boxplots)
Jupyter Notebook	latest	Interactive development environment

B. Preprocessing Pipeline

- Step 1: Load raw CSV → Strip quotes from column names and values
- Step 2: Rename columns with special characters (body fat_% → body_fat_pct)
- Step 3: Remove duplicate rows → Reset index
- Step 4: Cap body_fat_pct outliers at 95th percentile
- Step 5: Encode gender (M=1, F=0) and class (LabelEncoder: A=0, B=1, C=2, D=3)
- Step 6: Create 7 engineered features (BMI, pulse_pressure, fitness_score, etc.)
- Step 7: Feature selection (3-method ensemble → top 10 features)
- Step 8: Split data (80/20 primary + 70/30 + 50/50 for experiments)
- Step 9: Apply RobustScaler via Pipeline inside each model